Annotation guidelines for Gujarati dependency annotation based on AnnCorra and Paninian grammar

1. Introduction

1.1. Purpose

These annotation guidelines provide a systematic framework for the linguistic annotation of Gujarati sentences using a dependency-based approach grounded in Paninian grammar and the AnnCorra annotation framework. The guidelines are intended to ensure consistency and uniformity in the annotation of Gujarati syntactic and semantic relations.

The primary objective is to represent the relationships between words in Gujarati sentences by identifying the predicate/root,karaka relations,and other relevant dependency relations.

1.2 Scope

The guidelines cover the annotation of Gujarati sentences with particular emphasis on:

- Sentence tokenization
- Predicate and root identification
- Paninian Karaka assignment
- Other dependency relations
- Gujarati specific syntactic constructions
- Ambiguous construction cases
- Annotation validation and disagreement

The guidelines are based on the principles of AnnCorra and the dependency annotation framework developed for Indian languages, while incorporating decisions appropriate to the linguistic characteristic of Gujarati.

1.3 Guiding principles.

The following principles should be followed throughout the annotation process:

1. Consistency
   The same linguistic construction should be annotated using the same relation wherever possible.
2. Predicate/Root centred analysis:
   The main predicate serves as the central point for identifying the syntactic and semantic roles of the arguments.
3. Semantic consideration:
4. Karaka assignment should consider the semantic role of a constituent to a relation in the predicate and should not depend solely on surface word order or case marking.
5. Gujaratic specific adaptation:

The linguistic characteristic of Gujarati should be considered where the construction requires interpretation beyond the general AnnCorra guidelines.

6. Reproducibility:
   Annotation decisions should be sufficiently explicit so that the different annotations can apply the guidelines consistently

1.4 Annotation Workflow

The overall Annotation process follows
Gujarati sentences selection->Tokenization->predicate/root identification->karaka assignment->other dependency relations ->validation->disagreement analysis->final annotated sentences

The detailed rules for each stage are provided in further sections.

2. Tokenization guidelines

1. General tokenization rule

Each Gujarati sentences is divided into linguistically meaningful word units for annotation. Tokenization should preserve the original lexical units of the sentence.
The following conventions are followed:

1. Punctuation marks are not considered as token
   Commas,full stop, question marks, exclamation marks, quotation marks etc are excluded from annotation
   They are therefore not assigned a dependency relations or karaka label.
2. Abbreviations are not considered as separate tokens
3. Words are considered individual tokens.
   Each meaningful lexical word is assigned a token position.
4. Numbers are treated as tokens when they function as meaningful elements of the sentence
   For example,a year ,quantity or numerical expression participating in the sentence may be retained as a token
5. Proper names are treated as tokens according to their lexical form
6. The original order of the tokens is preserved.

3 . Predicate and Root identification

The predicate is the central element of a sentence and forms the basis for establishing the dependency structure and assigning karaka relations.

3.1 Identifying the main predicate

1. Identify the main predicate of the sentence before assigning other dependency relations
2. In a simple verbal sentence, the main verb is generally identified as the Predicate.

3. The Predicate should represent the primary action, event or state expressed by the sentence.
4. Once identified, the nation predicate is assigned the root relation
5. Other words in the sentence are connected directly or indirectly to the Predicate through dependency relations.
   Example
   રામ શાળાએ ગયો.
   Here
   ગયો is main predicate and root
   The remaining words are connected to the predicate according to their syntactic and semantic roles

3.2 Predicate in the Compound/Vector verb constructions

Gujarati frequently uses combinations of a main lexical verb and a vector/auxiliary verb.

For such type of constructions

- Identify the main verbal construction rather than treating every verbal element as an independent sentence root.
- The appropriate predicate structure should be determined based on the meaning and grammatical function of the verbs.
- Only one root should be assigned to a sentence.

3.3 Copular constructions

In sentences expressing identity, classification of a state using Copular construction, the central predicative element should be identified according to its role in the sentence. The Annotation should remain consistent across similar constructions.

3.4 Auxiliary verbs

Auxiliary verbs should not automatically be treated as the root merely because they occur after the main lexical verb. The annotator should first determine which verbal element carries the principal predicate meaning and then establish the dependency between the lexical predicate and Auxiliary.

3.5 Multiple verbal elements

When a sentence contains multiple verbal elements, the annotator should determine whether they form

- a single verbal construction
- A main verb with an auxiliary
- A compound/vector been construction
- Storage predicate in a coordinated/**computers** construction.

The same decision should be applied consistently to similar sentences.

3.6 Root selection rule

Each independent sentence should have on root

The root should be the central predicate from which the dependency structure can be established.

4 Dependency Assignment

Dependency Annotation represents the syntactic relationship between words in a sentence .Each token is connected to another token through a dependency relations thereby forming the dependency structure of the sentence.

4.1 General rules

1. The main predicate of the sentence is identified as the Root.
2. Every token other than the root should be assigned an appropriate dependency relations
3. The dependency relation should represent the grammatical relationship between dependent token and it's great.
4. Gujarat word order should be considered because the language permits relatively flexible ordering of the constituents.
5. The dependency structure should be constructed consistently across similar sentence constructions.

4.2 Predicate and its dependents

The main predicate acts as the central element of the Dependency structure. Its arguments and modifiers are connected to it through appropriate dependency relations

For example

રમેશે પુસ્તક વાંચ્યું.

The basic dependency structure is

```
      વાંચ્યું
      /    \
   રમેશે    પુસ્તક
```

Here

વાંચ્યું root

રમેશે dependent of વાંચ્યું

પુસ્તક dependent of વાંચ્યું

4.3 Dependency of modifiers

A modifier should be attached to the word that it directly modifies

For example

સુંદર ઘર

Here: ઘર → head

સુંદર → dependent of ઘર

સુંદર → adjmod

Thus the dependency relation represents the syntactic relationship between adjective and the noun.

4.4 Dependency of auxiliary and verbal elements

When a sentence contains an auxiliary or multiple verbal elements, the dependency relationship should be determined according to the grammatical structure of the verbal substring l construction.

4.5 Coordination

In coordinated constructions, the dependency structure should distinguish the main conduct from the additional conjuncts.

4.6 Consistency in Dependency assignment

The same dependency relation should be used consistently for similar syntactic constructions throughout the corpus. When an ambiguous construction is encountered, the annotator should refer to the rules and examples provided in these guidelines before making a decision.

4.7 Relationship with karaka annotation

Dependency Annotation and karaka annotation represent different aspects of the sentence structure

- Dependency relation describes the syntactic relationship between a dependent and its head
- Karaka relations represent the syntactico semantic role of an argument with respect to its root.

5. Karaka assignment guidelines

Karaka relations represent the semantic syntactic relationship between a constituent and the Predicate. The karaka should be assigned after identifying the root and establishing a proper relationship.

The following karaka labels are used in the annotation

Label	**Karaka**	**Interpretation**
K1	Karta	Agent/doer
K2	Karma	Object/participant affected by the action
K3	Karana	Instrument/means
K4	Sampradana	Recipient/beneficiary
K5	Apadana	Source/separation
K6	Sambandha	Possession/relation
K7	Adhikarana	Location
K7P	Place	Specific place
K7T	Time	Specific time

5.1 K1 -Karta
K1 is assigned to the participant that performs, industries or is responsible for the action expressed by the predicate
રમેશે પુસ્તક વાંચ્યું.
Ramesh read a book
Here
વાંચ્યું is the root
રમેશે is K1
પુસ્તક is K2
The participant રમેશે performs the actions of reading and therefore assigned K1.

5.2 K2 karma
K2 is assigned to the participant that is directly affected by, undergoes or is the target of the action expressed by the Predicate.
Example
રમેશે પુસ્તક વાંચ્યું.
Here in this statement ,પુસ્તક is the entity being read
રમેશે K1
પુસ્તક K2
વાંચ્યું Root
5 3 k3 karana
K3 represents the instrument,means or manner through which an action is performed.
Example
રમેશે છરીથી ફળ કાપ્યું
Here in this statement
રમેશે K1
છરીથી K3
ફળ K2
કાપ્યું Root
The છરી(knife) is the instrument used for performing the action.
Assign k3 when a constituent denotes the instrument or means used to perform the action.
5.4 k4 Sampradana
K4 represents the recipient, beneficiary or intended receiver associated with the predicate.
Example
રમેશે રીનાને પુસ્તક આપ્યું.
રમેશે K1
રીનાને k4
પુસ્તક k2

આપ્યું root
રીનાને is the recipient of the book and therefore receives the k4 relation.
Assign k4 when a constituent represents the recipient or beneficiary towards whom the action is directed.

5.5 K5 Apadana

K5 represents the source, starting point or the point of separation associated with an action or event

Example

રમેશ અમદાવાદથી આવ્યો.

રમેશ K1

અમદાવાદથી K5

આવ્યો root

5.6 K6 Sambandha

K6 represents the relation or possession association between two Nominal elements

રમેશનું પુસ્તક

રમેશનું K6

પુસ્તક K2

5.7 K7 Adhikarana

K7 is divided into two subtypes K7p and K7t .

5.7.1 K7p -location/place

It represents the location,place associated with an event or action

Example

રમેશ શાળામાં ભણે છે.

રમેશ K1

શાળામાં K7p

ભણે root

છે auxiliary

The school represents the location associated with the action

Assign K7p when a constituent specifies the place in which an action or event occurs

5.7.2 K7t Time/Temporal

K7t is used when a constituent specifies the time or temporal period associated with an action or event.

Example

રમેશ સવારે શાળાએ ગયો

Here

રમેશ is K1

સવારે K7t

શાળાએ K7p

ગયો root

Assign K7t when the constituent specifies when an action or event takes place

K7 decision rule

If the constituent answers “Where?” K7p

If the constituent answers “When?” K7t

6 Other dependency relations

Not all tokens in a sentence represent Paninian Karaka relations.Some tokens function as auxiliaries,coordinators or other syntactic elements. These relations are represented using appropriate dependency labels.

The following guidelines should be followed for assigning non karaka dependency relations

6.1 Nominal subject(nsubj)

The nsubj relation is used when a nominal element functions as the syntactic subject of a predicate but is not assigned a karaka label in the annotation scheme

6.2 Object(obj)

The obj relation is used for an object when the corresponding dependency representation requires an object relation rather than a Paninian Karaka label.

6.3 Adjective modifier(adjmod)

adjmod is assigned when an adjective modifies a noun.

Example

મોટું ઘર

Here મોટું modifies ઘર and it is assigned adjmod

6.4 auxiliary (aux)

An auxiliary verb is attached to the main predicate using the appropriate auxiliary relation.The main lexical predicate remains the central element of the verbal construction

6.5 Coordination (cc)

In coordinated sentences cc is used for the coordinating conjunction.

The Annotation should consistently identify the principal conjuct and attach the additional conjunct accordingly

6.6 Reference relation (rs)

Reference relations (rs) is used when a phrase or expression refers to ,relates to or makes reference to a previously mentioned or contextually understood entity ,matter or event.

7 Gujarati specific annotation decisions

This section defines annotation decisions for constructions that are particularly relevant to Gujarati.The aim is to ensure that Gujarati syntactic and semantic characteristics are represented consistently while following AnnCorra and Paninian framework.

7.1 Postpositional expressions

Gujarati commonly used postpositions to express grammatical and semantic relations.Annotators should consider the complete Postpositional expressions when determining its function.

7.2 locative expression K7p
Expressions indicating place or location should be assigned K7p
Example
બાળકો બગીચામાં રમે છે
Here in this statement, બગીચામાં answers the question where “Where?”
Annotation
બગીચામાં K7p
7.3 Temporal expression K7t
Expressions introducing time or temporal information should be assigned K7t
Example
રોહિત ગઈકાલે મુંબઈ આયવો
Here in this statement ગઈકાલે answers the question “When?”
Annotation
ગઈકાલે K7t
7.4 Reference Expression rs
Expressions that refer to a previously mentioned or contextually understood matter ,entity or topic should be assigned rs when they do not function as karaka relations.
Example:
તેમણે આ અંગે વધુ વાત કરવાનો ઇનકાર કર્યો.
Here આ અંગે refers to a previously mentioned matter.
Annotation:
આ અંગે is rs
7.5 Flexible word order
Gujarati allows relatively flexible word order, particularly for emphasis and discourse purposes. Therefore dependency Annotation should not be based on only the linear position of a word.
Rule: The syntactic semantic function of a constituent should determine its admission, not its position in the sentence.
7.6 Coordination
The conjunction અને is annotated as the coordination relation when it connects two or more Coordinated constituents
Example
રવિ અને મયુર ફરવા ગયા
Here રવિ and મયુર are the coordinated elements
અને functions as the coordinating conjunction
When a coordinating conjunction such as અને connects two constituents, the conjunction itself should be assigned the coordination dependency relation
7.7 Consistency across Gujarati constructions
For all Gujarati specific constructions

- The same construction should receive the same annotation wherever possible

- Surface word order should not determine the label by itself
- Connect should be considered when resolving Agnihotri
- Karaka labels should be assigned according to semantic syntactic function
- K7p and K7t should always be distinguished according to place and time respectively.

8 Annotation Validation and quality control

8.1 Purpose of Validation

Validation is performed to identify annotation errors, inconsistencies and disagreements between annotators . It helps ensure that the Annotation guidelines are applied consistently across the corpus

8.2 Independent annotation

A subset of the annotated sentences should be independently reviewed by a second annotator familiar with Gujarati grammar and the Annotation framework.

The annotators should independently assign

- Predicate/root
- Dependency relatives
- Karaka relations
- Other dependency relations

The independent annotation should be performed without influencing the second annotator through the first annotator's decisions.

8.3 Inter Annotator agreement

The agreement between annotators should be measured to evaluate annotation consistency using measures such as percentage agreement and Cohen’s Kappa

8.4 Disagreement Analysis

Disagreements should be examined to identify the source of inconsistency. Particular attention should be given to ambiguous Karaka assignment,predicate identification and coordination.

8.5 Guideline Refinement

If recurring disagreements are identified the relevant annotation rules should be classified or refined to improve consistency.

8.6 Final quality check

The annotation should be checked for missing relations, inconsistent labels and violations of the defined annotation convention before finalizing the corpus

Conclusion

The Annotation guidelines provide a systematic procedure for Gujarati dependency shoaib based on AnnCorra and Paninian grammatical framework. The guidelines establish conventions for tokenization, predicate identification, dependency assignment,karaka relations,ambiguity resolution and annotation consistency.

The guidelines serve as a practical reference for annotators and help ensure that the resulting Gujarati annotated resource is linguistically consistent, reproducible and suitable for further NLP research.